Time Series Forecasting for PGCB Electricity Demand

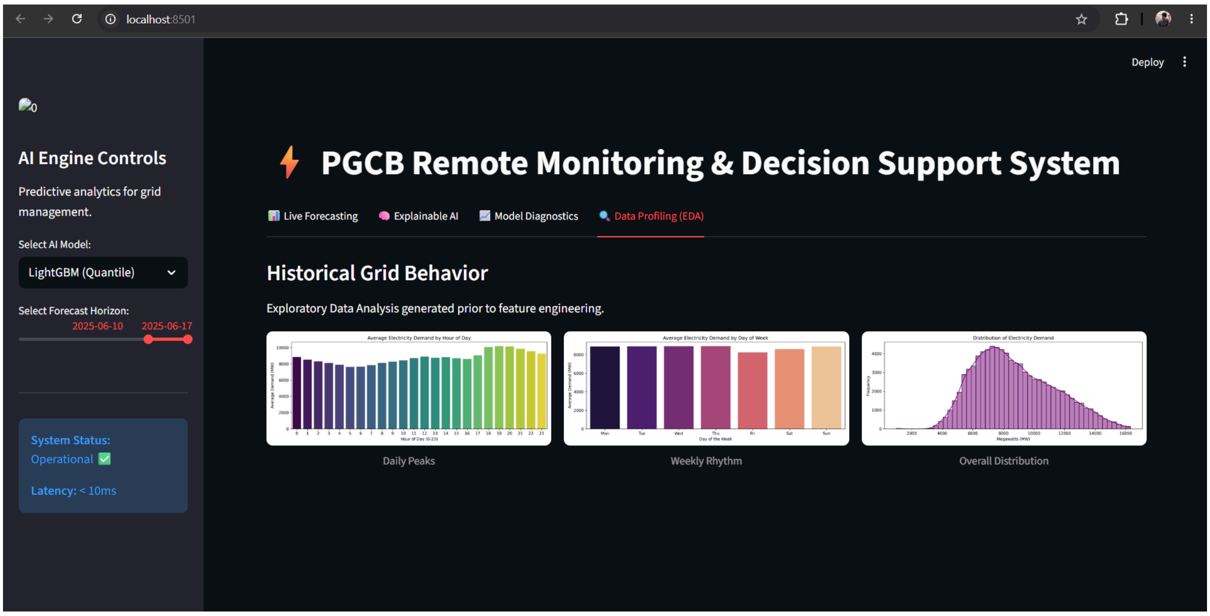
Introduction

This report outlines the end-to-end development of an AI-powered electricity demand forecasting pipeline and interactive web dashboard. Using the PGCB Hourly Generation Dataset, the objective was to predict national electricity demand and serve the insights through a Decision Support System. By engineering robust temporal features and introducing an innovative Quantile Regression model, the final system achieved a Mean Absolute Percentage Error (MAPE) of 1.59%, establishing a highly reliable tool for grid management.

Data Preprocessing & Feature Engineering

To prepare the dataset for time-series forecasting, strict sequential integrity was prioritized.

- Resampling & Interpolation: The raw data contained uneven intervals (e.g., half-hour records). The target variable was resampled to strict 1-hour intervals, and missing values were handled using time-based interpolation to maintain a continuous, gap-free sequence because using global mean, mode or median imputation was strictly avoided, as inserting a dataset-wide average would create artificial spikes or drops, destroying the local chronological sequence.
- Outlier Treatment: The Interquartile Range (IQR) method was applied. Extreme spikes were clipped rather than dropped to prevent disrupting the time-series continuity.
- Feature Engineering:
    - Calendar & Cyclicality: Extracted the hour, day, and month. Cyclical encoding (sine-cosine transformations) was applied to the hour variable to ensure the model mathematically links 23:00 to 00:00.
    - Autoregressive Lags & Rolling Stats: Generated historical lags (1 hour, 2 hours, 24 hours, and 168 hours) to capture immediate and weekly temporal dependencies. A 24-hour rolling mean and standard deviation were also calculated, using a shift to prevent data leakage. By strictly applying a one-step shift to these rolling statistics, the model was forced to calculate averages using only past data, completely eliminating the risk of the algorithm artificially 'peeking' at the current hour's target variable during training.



Modeling Strategy & Performance Analysis

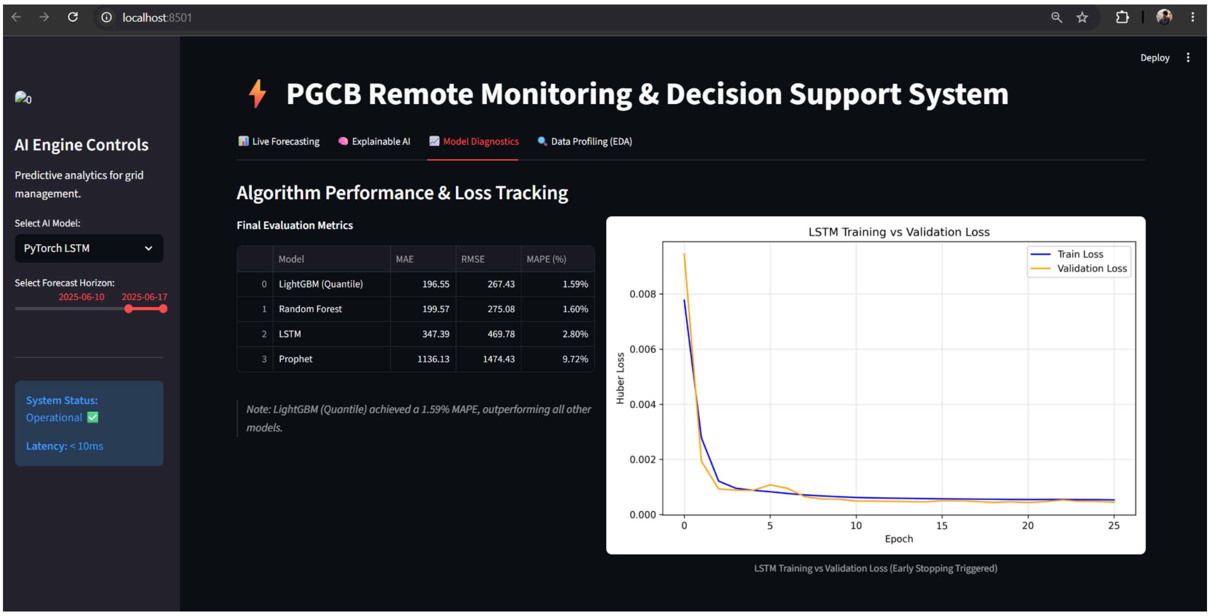
To evaluate algorithmic efficacy, four distinct models were trained to predict electricity demand one hour ahead. A strict chronological train/validation/test split was enforced (the final 30 days were reserved for testing), ensuring no future data leaked into the training phase.

- Pure Seasonal Baseline (Facebook Prophet): Trained purely on datetime and target variables to establish a cyclical baseline.
- Classical ML (Random Forest): Tuned using TimeSeriesSplit validation to capture non-linear relationships via the engineered lag features.
- Deep Learning (PyTorch LSTM): Built specifically for sequential data. The architecture utilized Huber Loss for outlier robustness and Early Stopping based on chronological validation set to prevent overfitting.
- Champion Model (LightGBM Quantile Regression): An advanced tree-based model trained using Pinball Loss to predict the 10th, 50th (median), and 90th percentiles simultaneously.

Final Model Evaluation Metrics:

Model	MAE	RMSE	MAPE
Random Forest	199.57	275.08	1.60%
PyTorch LSTM	347.39	469.78	2.80%
Facebook Prophet	1136.13	1474.43	9.72%
LightGBM (Quantile)	196.55	267.43	1.59%

The baseline Prophet model yielded a 9.72% error, proving that pure seasonality is insufficient. The integration of autoregressive lag features allowed both Random Forest and LightGBM to drastically reduce error to ~1.60%. LightGBM was selected as the final production model due to its superior accuracy, computational efficiency, and ability to generate mathematical confidence bounds. Prophet is fundamentally a univariate model designed for seasonal trends; it underperformed because its architecture could not ingest our highly predictive engineered features (like lag_1), proving that recent physical context is far more critical than pure calendar seasonality for grid forecasting. Furthermore, while PyTorch LSTM is powerful for complex sequential data, LightGBM and Random Forest outperformed it because tree-based algorithms are highly efficient at splitting on dominant autoregressive features in tabular time-series data without requiring the massive historical data volumes that neural networks typically need to converge.



Hyperparameter Optimization & Cross-Validation

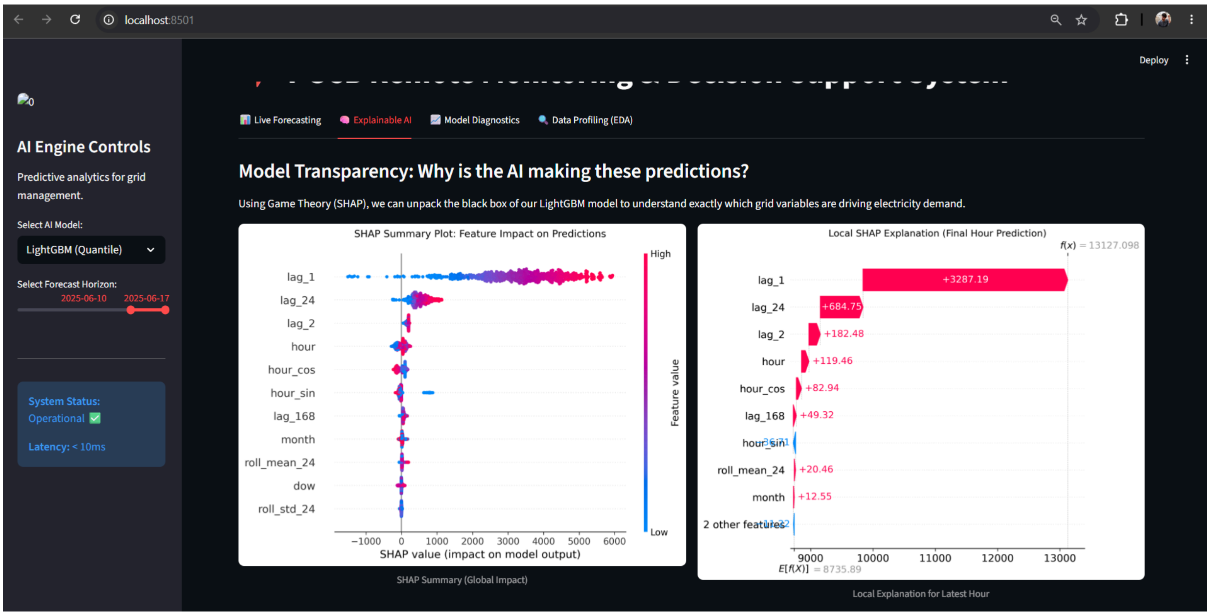
To prevent future data leakage, TimeSeriesSplit (forward-rolling validation) was strictly utilized over standard randomized cross-validation.

- Random Forest: Optimized to n_estimators=200, max_depth=15, and min_samples_split=2. Constraining tree depth was critical to prevent the model from memorizing historical anomalies.
- PyTorch LSTM: Regularized via a 20% dropout rate. Training was dynamically halted at Epoch 26 via Early Stopping when validation loss plateaued, explicitly preventing the neural network from overfitting.
- LightGBM (Quantile): Tuned for stability (learning_rate=0.05, max_depth=10, n_estimators=300). The alpha hyperparameter was explicitly set to 0.1, 0.5, and 0.9 to optimize the Pinball Loss function for true statistical bounding.

Explainable AI (XAI) & Model Interpretability

A core requirement of a robust Decision Support System is transparency. To unpack the LightGBM model's decisions, Game Theory (SHAP - SHapley Additive exPlanations) was implemented.

- Global Interpretability: The SHAP summary plot confirmed that the most recent historical observation (lag_1) is the strongest driver of future demand, followed by cyclical daily patterns.
- Local Interpretability: Local waterfall plots were generated to explain exactly which variables pushed the model's prediction higher or lower for any given specific hour.

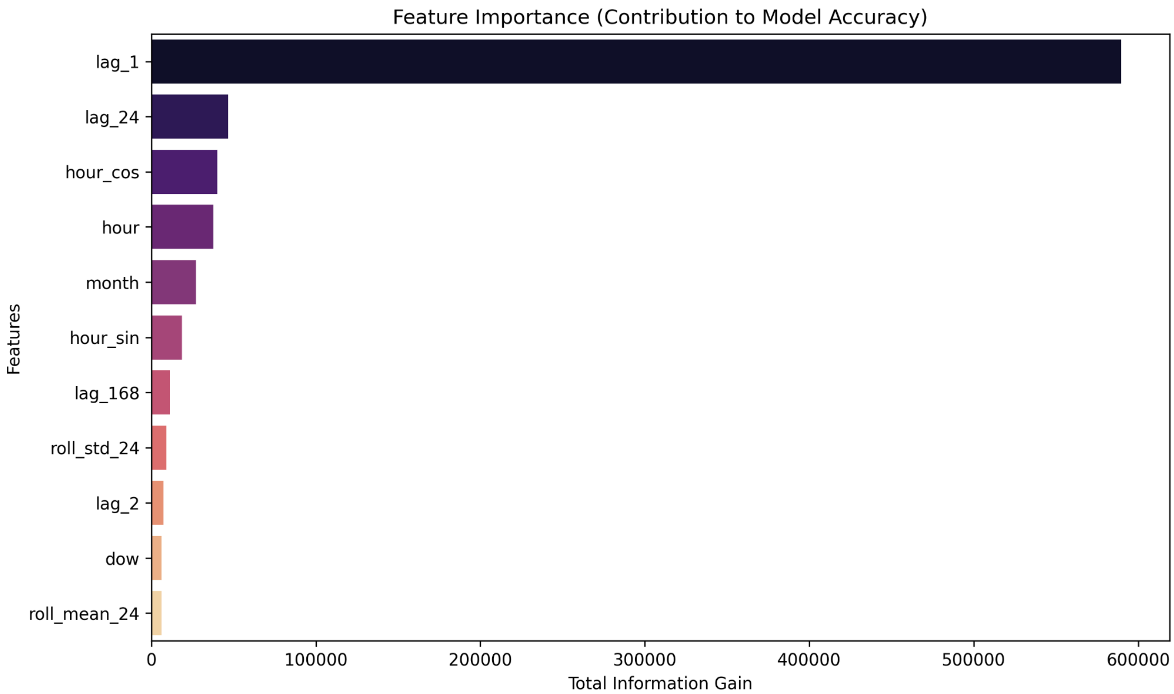


3.1 Feature Importance (Information Gain)

Information Gain was extracted from the LightGBM model to measure each feature's direct contribution to error reduction, revealing a clear hierarchy of grid drivers:

- Autoregression (lag_1): Dominated the importance metrics. Grid loads possess high physical inertia; the preceding hour's demand is the strongest statistical predictor of the current hour, overriding general seasonality. Because of this high physical inertia, sudden macro-events, such as a massive storm causing a rapid drop in temperature will immediately reflect in lag_1, allowing the LightGBM model to instantly course-correct its next prediction where a purely seasonal model would fail.
- Rolling Statistics (roll_mean_24): Acted as a critical secondary feature. It establishes a dynamic daily baseline, allowing the model to adapt to macro-shifts like multi-day weather events.

- Cyclical Calendar Features (hour_sin/cos): Successfully allowed the model to mathematically differentiate between stable overnight base-loads and highly volatile evening peaks.

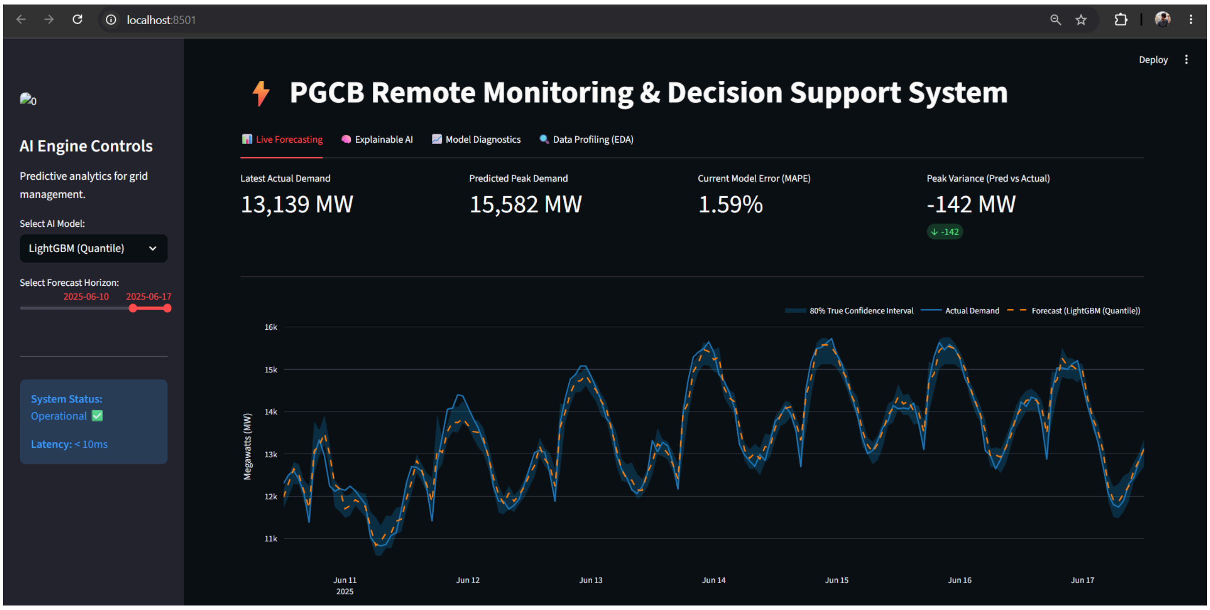


Interactive Dashboard & Decision Support System

A comprehensive web dashboard was developed to serve the AI insights to end-users. Built using Streamlit and Plotly, the multi-tab interface exceeds basic requirements by offering live forecasting alongside model diagnostics and XAI.

- Dynamic Visualizations: The primary interface renders a time-series graph of historical data (solid blue line) overlaid with the forecasted future values (dashed orange line).
- Mathematical Confidence Bands: Because the LightGBM model utilizes Quantile Regression, the dashboard plots a mathematically true 80% Confidence Band (the gap between the 10th and 90th percentiles) rather than arbitrary shading. From a grid management perspective, the 90th percentile upper bound is critical for risk mitigation; it defines the 'worst-case scenario' peak demand, allowing operators to proactively schedule spinning reserves and prevent national blackouts.

- Summary Panel & Interactive Controls: A KPI panel summarizes vital statistics, including the latest actual demand (13,139 MW), the predicted peak (16,003 MW), and the live model error variance. Users can manipulate the data using interactive controls, including a model selector dropdown and a forecast horizon date slider.



Limitations

- Absence of Exogenous Data: The current pipeline relies entirely on endogenous variables (historical load and time). It does not factor in meteorological data (e.g., temperature, humidity), which is a primary driver of HVAC-related electricity demand.
- Geographic Granularity: The model forecasts national aggregate demand. It cannot identify localized grid stress or regional transformer bottlenecks.

Future Scope

- Weather API Integration: Ingesting live weather forecasts would significantly improve the model's ability to predict unseasonal demand spikes caused by sudden heatwaves or cold snaps.
- Deep Learning Advancements: Future iterations could explore Temporal Fusion Transformers (TFT) or N-BEATS architectures, which are natively designed to process multiple exogenous time-series variables simultaneously.